

BlazeSync — Master PRD

Ecobank InnovateX 2026 | Team Veyra | Inclusive Finance Track

This document supersedes earlier drafts and consolidates survey data, the Course Advisor mandate model, the virtual-account architecture, the AI layer, and the monetization model into one source of truth.

1. Executive Summary

BlazeSync is an institutional financial transparency platform for university student associations. Every member gets a unique virtual account; payments are auto-reconciled the instant they land — no screenshots, no manual matching. Every disbursement requires multi-signature approval, with an optional Course Advisor sign-off layer for institutional accountability. Every member gets an automatic, verifiable receipt that also tells them exactly what their money is funding — closing the loop between "I paid" and "I know what I'm getting." The whole system is built on Ecobank's Unified-API, turning recurring association dues into a youth account acquisition and deposit-float engine for Ecobank.

2. Problem Statement (Survey-Backed)

The moment, told from both sides:

At a general meeting, a treasurer tries to explain where the semester's dues went. A member asks for proof. The treasurer pulls up a WhatsApp chat full of screenshots. Nobody in the room — including the treasurer — can say with certainty what the current balance actually is.

For students, this is where trust breaks down. From our own survey of 17 students across multiple levels:

- **88.2%** have personally suspected, witnessed, or heard about dues mismanagement or missing funds
- **58.8%** rate clarity on how dues are spent as a 1 or 2 out of 5
- A majority report general meetings devolving into arguments over "where the money went"
- Open-ended responses, unprompted: *"lack of transparency about how dues are collected, what they are used for", "not having proper documentation of how the money was used", "we don't see the benefit"*

For treasurers and exco, this is where they become unfair suspects. **52.9%** of students say proof of payment today is a bank transfer screenshot on WhatsApp — unverifiable for the

student, indefensible for the treasurer who has no fast, credible way to prove innocence when accused.

Both sides want the same missing thing: a single, unquestionable source of truth. 58.8% of respondents called a shared live digital ledger "extremely valuable," with nearly all the rest saying "somewhat valuable."

(Sample caveat for internal use: n=17, skewed toward 400/500 level — self-selected respondents likely care more than average. The direction of the finding is still strong and consistent across multiple questions, and matches a pattern already validated by a past hackathon winner — GTCO's "Team Within" — solving the same core problem.)

3. Solution Overview

BlazeSync replaces the WhatsApp screenshot — for everyone, at the same time:

- **Virtual account per member** — payments are automatically attributed the instant they land, no manual reconciliation
 - **Live, append-only ledger** — visible to exco always, visible to members on demand, never requiring active engagement to be meaningful
 - **Multi-signature disbursement approval** — no single person can move money alone
 - **Optional Course Advisor sign-off** — institutional mandate layer, see Section 5
 - **Automatic hashed receipts** — tamper-evident proof of every payment
 - **Expectation Receipt** (new, from PM's research) — every dues cycle, exco states upfront what the money is funding ("Dept Dinner, exam prep sessions, welfare support"); this is shown to the student before/at payment and attached permanently to their receipt. This directly answers the survey's loudest complaint — *"we don't see the benefit"* — without requiring the student to go dig through a ledger at all. It sets an explicit, checkable promise, and gives exco accountability even to the students who will never open a transaction feed.
 - **AI layer** — see Section 7
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4. Stakeholder Matrix

Stakeholder	Role	Core pain solved	Financial relationship
Student EXCO Body (<i>primary buyer</i>)	Operational admin & budget manager	Eliminates manual reconciliation, accusation risk, accounting chaos	Pays platform subscription + micro-fees from association funds
Course Advisor / Patron (<i>governing body</i>)	Compliance officer, financial approver	Removes personal liability and audit friction via line-item visibility	Indirect — mandates BlazeSync for EXCO sign-off, free portal access
Students (<i>payers</i>)	Members & contributors	Payment verification + visibility into what dues fund	Zero cost — pay standard dues only, no surcharge
Ecobank (<i>banking partner</i>)	Infrastructure provider	Drives transaction volume, acquires deposit float and youth accounts	Net revenue — transaction fees, float yield, CAC reduction

5. Why Adoption Doesn't Depend on a Corrupt Treasurer's Goodwill

A fair, hard question: what stops a finsec who actually wants to commit fraud from simply not adopting BlazeSync?

Two real answers, not just hope:

1. **For the honest majority** (most finsecs — people doing an unpaid, thankless job who get accused unfairly), BlazeSync is protection, not surveillance: *"I use a system where I literally cannot move money alone, and everything is logged — don't take my word for it, check it yourself."* That's a strong voluntary incentive.
2. **For institutional-level adoption**, the Course Advisor/Patron role is the real lever: when a faculty advisor — who already holds formal authority over association accounts — mandates BlazeSync for sign-off and fund release, adoption stops being any individual treasurer's personal choice. It becomes a condition of holding the account at all, the same way an advisor already has to countersign in most real departments.

We don't claim to eliminate fraud outright — no system does. We make transparency the default, easy choice for the honest majority, and make opting out of it a visible, costly signal for anyone who won't.

6. Business Model

Who pays: the Student EXCO Body, as a B2B buyer controlling central budgets (ranging ₦500,000–~~₦15,000,000~~+ per session). Software cost is budgeted as an official "Administrative & Software Overhead" line item. Students themselves pay zero added cost — standard dues only.

Revenue mechanics:

1. **Transaction platform fee** — 1.5% micro-fee, capped at ₦200 per transaction, embedded in collection or auto-deducted from total dues collected
2. **Tiered session subscription (SaaS):**
 - Tier 1 (small depts, <200 students): ₦15,000/session
 - Tier 2 (large faculties, >1,000 students): ₦45,000/session — includes custom audit exports
3. **Sponsorship placement fee** (*Vision, not MVP*) — corporate brands pay to access student event proposals via a sponsor module

Ecobank's revenue, as the banking partner:

1. Transaction interchange / NIP processing fee splits (~~₦3.75~~–₦50 per transfer) on incoming/outgoing virtual account traffic
 2. Float deposit yield — associations retain significant balances in virtual vaults for weeks/months ahead of events; Ecobank monetizes this float
 3. Low-CAC institutional & youth customer acquisition — EXCO-mandated corporate accounts, plus a funnel for graduating students into personal Ecobank/Blaze accounts
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7. AI Layer

Not decorative — each piece directly answers a hard question the product otherwise can't answer on its own.

MVP — built and demoed live:

- **Ask BlazeSync** — a member or exco can ask in plain language ("how much did we spend on the dept dinner?", "has my level's levy been accounted for?") and get a direct answer pulled from the live ledger. This is the real fix for the survey finding that most students won't actively dig through a transaction feed — transparency stops requiring effort.

Vision — roadmap, stated clearly as not-yet-built:

- **BlazeSync Sentinel** — an autonomous agent that continuously watches disbursement patterns and proactively flags anomalies without being asked: unusually large disbursements relative to an association's history, multiple smaller disbursements that

look like they're being split to dodge the multi-sig threshold ("structuring"), spending spikes right before a handover. This is the technical answer to Section 5's fraud question — a complement to the social/institutional answer, not a replacement for it.

8. Features — MVP (Built) vs. Vision (Roadmap)

MVP — what's actually demoed

- Treasurer/exco setup, roster upload, per-member virtual account generation
- Invite flow (claim virtual account + app access via single-use code)
- Dues cycle creation, including the **Expectation Receipt** statement
- One-tap payment → instant auto-reconciliation → hashed receipt with expectation statement attached
- Live append-only ledger, visible to exco and members
- Disbursement request → multi-signature approval (server-enforced) → payout → ledger update with reason
- **Ask BlazeSync** (AI, working)
- Balance reconciliation against Ecobank's Account Enquiry (our ledger vs. their source of truth)

Vision — stated as roadmap, not demoed

- Course Advisor / Compliance Portal: disbursement voucher review with document previews, line-item approve/flag/reject, digital signatures, full audit package export
- Dynamic Budget Builder (line-item budgets with vendor quotes/attachments)
- Corporate Sponsor Tracking module (dedicated funds ledger, proof-of-execution uploads, sponsor impact reports)
- BlazeSync Sentinel (AI anomaly detection)
- Multi-association dashboards for Students' Unions managing several departments at once

Pitch framing: *"What you're seeing today is our MVP — it proves the core mechanism works. Here's where this goes next as a full institutional platform."* Say the line out loud. It's the difference between "ambitious vision" and "vaporware," which judges are explicitly trained to penalize.

9. End-to-End User Journeys

Journey 1 — Dues Collection & Verification Student pays into their unique virtual account → Ecobank webhook fires payment confirmation → BlazeSync auto-reconciles the ledger and

issues a hashed receipt (with the Expectation Receipt statement attached) → exco's register updates in real time.

Journey 2 — Disbursement & Approval Treasurer creates a disbursement request (amount, reason, recipient) → exco co-signatories review and approve/reject → once the signature threshold is met → Ecobank releases funds → ledger updates with an outflow entry tagged with the reason → all members notified of what was spent and why.

(Vision, not MVP: Course Advisor review step inserted between exco approval and fund release, with document preview and digital signature.)

10. System Architecture (Summary)

- **Frontend:** Next.js (React) + Tailwind — Treasurer/Exco Portal, Member Portal
 - **Backend:** FastAPI (Python), PostgreSQL, Redis, Celery/BackgroundTasks for async jobs
 - **Ecobank integration:** Authentication Service, Collection Service (virtual accounts), Account Enquiry Service, Local Bank Payment Service, Notification Service (webhooks)
 - **Security:** JWT auth with RBAC, server-side enforced multi-sig threshold, append-only ledger, deterministic recomputable receipt hashes, verified webhook signatures, immutable audit log
 - *(Full data model, endpoint list, and security rationale live in the separate Technical Build Spec and Backend Build Prompt documents — this PRD is the narrative/product source of truth, those are the implementation source of truth.)*
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11. Operational Success Metrics (KPIs)

- **Approval cycle time:** reduce budget/disbursement turnaround from days to under 2 hours
 - **Reconciliation accuracy:** 100% automated matching of dues payments, zero manual screenshot processing
 - **Adoption rate:** target 80%+ adoption among departmental associations within partner universities in Year 1
 - **Financial trust index:** target average student trust score above 4.0/5.0 on end-of-session surveys (baseline from our own survey: clarity currently sits at 1-2/5 for 58.8% of respondents)
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12. Pitch Structure

1. **Hook** — the shared-incident problem statement (Section 2), told from both sides
2. **Live demo** — the MVP golden path (Section 8), uninterrupted
3. **Ecobank synergy** — transaction fees, float yield, CAC reduction (Section 6)

- 4. **Business model** — EXCO-pays structure, tiered subscription (Section 6)
 - 5. **The hard question, answered before it's asked** — "what stops a corrupt treasurer from avoiding this" (Section 5)
 - 6. **Vision slide** — MVP vs. roadmap, stated explicitly (Section 8)
 - 7. **Close** — adoption targets and trust-index goal (Section 11)
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13. Key Dates

Milestone	Date
Regional Stage — Yaba College of Technology	9 October 2026
Regional Stage — University of Port Harcourt	15 October 2026
Regional Stage — University of Abuja	20 October 2026
Grand Finale — Ecobank Pan African Centre, VI, Lagos	30 October 2026